ModelIt!

Student Activity Pack

The Mushroom Network Under Your Feet

Trees Share Food and Warnings Through Underground Internet

Grade 5 | NGSS 5-LS2-1 | Systems and System Models

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

© 2026 Discovery Collective. All rights reserved.

Table of Contents

ModelIt!

1 Pre-Assessment 3

2 Scoring Rubric 4

3 Component Analysis 5

4 Model Recording Page 6

5 Simulation Observations 7

6 Research & Extend 8

7 Reflection Questions 9

8 STEM Challenge 10

1 Pre-Assessment

ModelIt!

What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Do trees in a forest compete, cooperate, or both? Explain.

2 What do mushrooms/fungi do in an ecosystem?

3 If trees could share food underground, what would have to be doing the sharing?

4 What is the Wood Wide Web? Have you heard of it?

2 Scoring Rubric

ModelIt!

Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

ModelIt!

What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

- Leaf Litter / Organic Matter
- Soil Moisture
- Mycorrhizal Fungal Network Activity
- Nutrient Transfer Between Trees

EXTERNAL	INTERNAL

? Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

ModelIt!

Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

ModelIt!

Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Healthy Forest

Normal organic matter and moisture

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Drought Scenario

Lock Soil Moisture to 10%

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Deforestation

Lock Leaf Litter to 0%

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

ModelIt!

Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Soil pH — Acidic soil from acid rain damages fungi

Tree Species Diversity — Different species connect differently

Mother Tree Presence — Oldest trees are network hubs

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

ModelIt!

The Big Question

How do underground conditions control how much trees can share with each other? Use evidence from your simulation to support your answers.

1. **Why might a cleared forest take decades to recover even with replanting?**

2. **How does drought affect the entire forest community, not just individual trees?**

3. **What does this network tell us about competition vs. cooperation in nature?**

4. **How is the mycorrhizal network similar to and different from the internet?**

5. **Why are 'mother trees' so important to protect?**

8 STEM Challenge: Design Forest Recovery Plan

ModelIt!

YOUR ENGINEERING MISSION

Design a plan to restore the mycorrhizal network in a cleared forest.

THE SCENARIO

A forest was clear-cut 5 years ago. New trees are struggling. Design a plan to restore the underground network.

Engineering Design Thinking

1. WILL YOU USE FUNGI FROM NEARBY HEALTHY FORESTS?
2. HOW WILL YOU PROTECT NEW FUNGAL GROWTH?
3. WHAT ORGANIC MATTER WILL YOU ADD?
4. HOW LONG WILL FULL RECOVERY TAKE?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Mycologists and Forest Ecologists study the underground networks that hold forests together. They earn $55,000-$90,000/year.